CFD ENGINEERING

SAI VEERESH BANDI

CFD ENGINEER — COMPRESSIBLE & HYPERSONIC FLOWS

Mach 2–20 · ANSYS Fluent · OpenFOAM · Shock & Expansion Analysis

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- Delhi, India

ABOUT

Engineering at the edge of high-speed flow

I specialize in compressible and hypersonic flows (Mach 2–20), where shock waves, expansions and thermal effects dominate. Trained through Flow ThermoLabs' advanced CFD program, I build stable, physically accurate simulations in ANSYS Fluent and OpenFOAM — with an emphasis on solver strategy, high-quality meshing, turbulence modeling and rigorous validation against published data.

Driven by propulsion and ambitious aerospace systems — from deep-space exploration to fast-iterating launch startups — I thrive on curiosity in the details, and always validating.

Mach 2–20

Compressible → hypersonic regimes simulated

6+

Independent CFD case studies, mesh-to-validation

<0.1%

Mass-imbalance achieved on converged solutions

EXPERTISE

Skills & Tools

CFD & Physics

- Compressible & hypersonic flows (Mach 2–20)
- Shock wave & expansion analysis
- Boundary layer behavior, high-speed regimes
- Turbulence modeling (k-ω SST)
- Species transport & reacting flows

Numerical Methods

- Density-based & pressure-based solvers
- AUSM schemes
- Mesh & domain independence studies
- Convergence & residual analysis

Tools

- ANSYS Fluent · OpenFOAM
- MATLAB · Python (post-processing)
- SolidWorks · CATIA · HyperMesh

EXPERIENCE

Professional Experience

Design Engineer

HRBL Group · Nov 2024 – Present · Delhi, India

- Designed & analyzed EV chassis, battery housings, cooling systems and structural mounts (SolidWorks, CATIA)
- Led structural, modal, thermal & CFD-driven analyses in ANSYS to optimize durability and cooling efficiency
- Ran CFD thermal simulations for battery/motor cooling; reverse-engineered legacy designs to cut cost
- Performed multiphysics (structural + thermal + fluid) validation against performance & safety standards

Product Development Intern

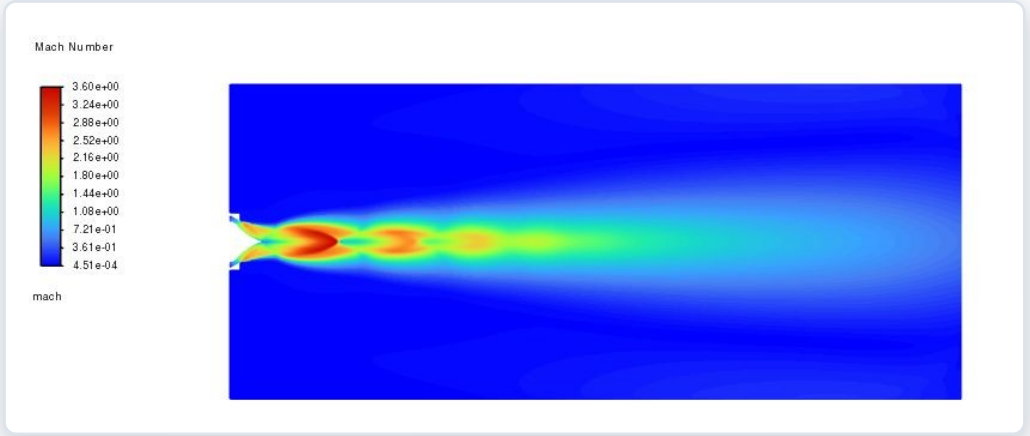
Simulation Lab · Sep – Oct 2024

- Conducted aerodynamic CFD analysis on subsonic aircraft wing geometries in ANSYS Fluent
- Computed lift/drag coefficients; refined meshes and boundary conditions for accurate force prediction

PROJECT 01

Aerospike Nozzle — Expansion Behavior

2D axisymmetric · Density-based solver · k-ω SST



- Modeled a modified 2D axisymmetric aerospike to study exit-plume expansion under compressible flow
- Nozzle relations sized the geometry before mesh generation and CFD analysis
- Mesh-convergence study across coarse / medium / fine grids (25k → 486k elements) to confirm solution independence

Inlet: 15.2 bar, 1180 K · Outlet: 101,325 Pa · Ideal-gas air, Sutherland viscosity · Fine mesh: 486k elements

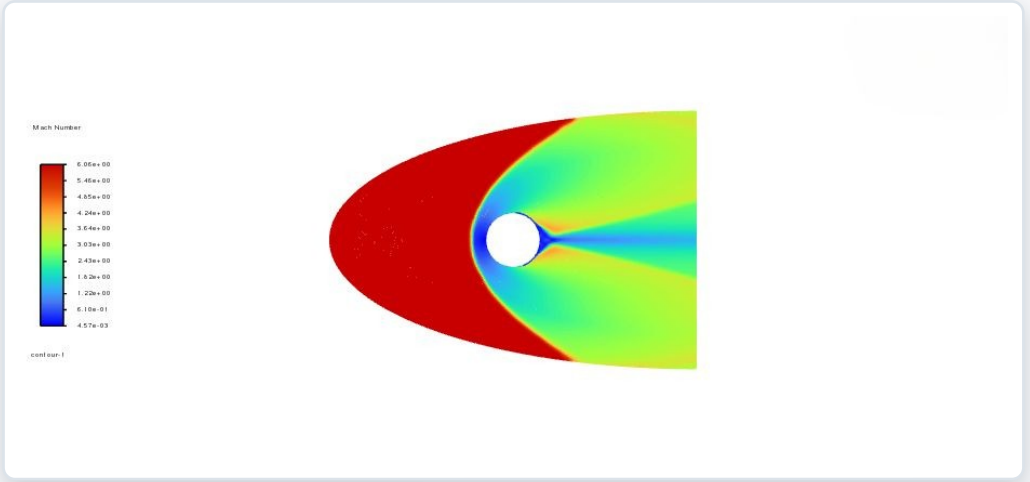
Results show clear shock-diamond formation from the exit pressure imbalance, with Mach number converging from 2.67 (coarse) to 3.90 (fine mesh) — confirming grid-independent capture of the periodic expansion/recompression structure.

PROJECT 02

Rotating Cylinder in Mach 6 Flow

Transient · Density-based, explicit AUSM · k-ω SST

- 75 mm cylinder spinning at 1,074 rad/s in Mach 6 freestream, to study rotation–shock coupling
- Quad-dominant mesh with 20 inflation layers (0.05 mm first cell) to resolve the boundary layer
- Transient, first-order implicit formulation to capture the evolving wake



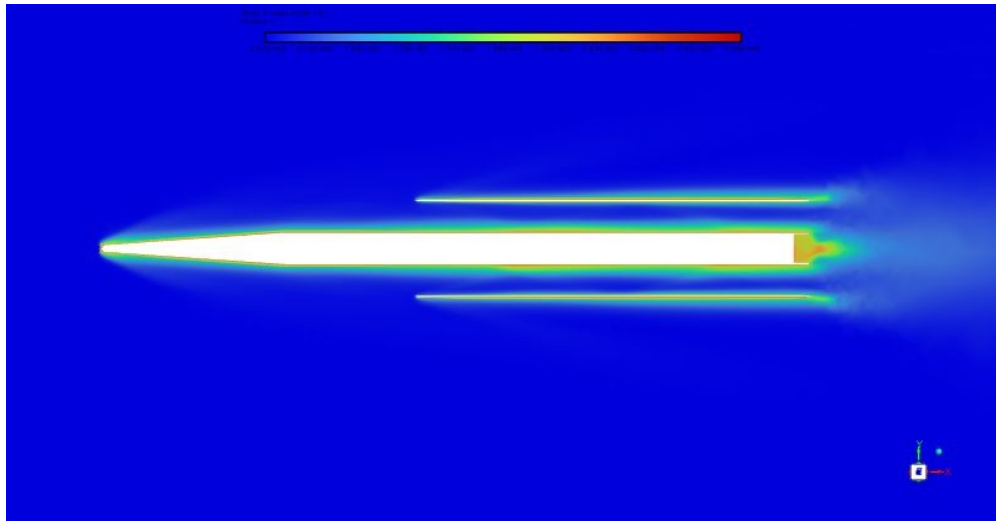
Freestream: 1,951 Pa, 58.25 K, Mach 6.06 · 34,079 elements · Growth rate 1.2

A detached bow shock forms ahead of the cylinder, and the high spin rate (1,074 rad/s) breaks flow symmetry — visibly skewing the pressure and velocity distribution around the surface.

PROJECT 03

NASA X-43 Hypersonic Flow Analysis

Mach 7 · Frozen chemistry vs. species transport



- Geometry sourced from GrabCAD and rebuilt for CFD: mesh prep, domain sizing, symmetry model
- Compared frozen-chemistry against multi-species air (N_2, O_2) to capture high-temperature gas effects
- NASA 9-piecewise polynomial Cp with Sutherland viscosity for accurate thermodynamics at Mach 7

Freestream: 1,400 Pa, 226 K, Mach 7 · ~1.45M cells (polyhedral core + 15-layer prism boundary layer)

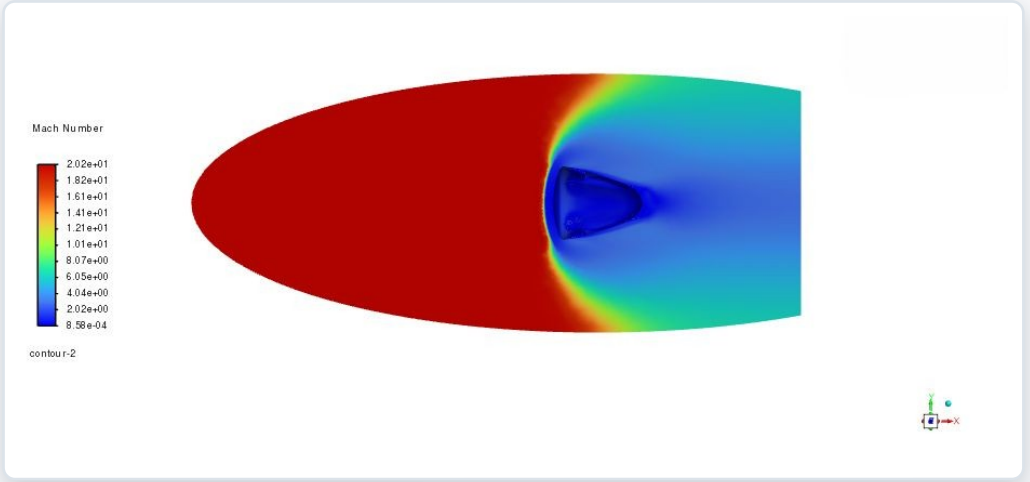
Parameter	CFD (this work)	Published
Drag coefficient (Cd)	0.0166	0.015 – 0.018 [1,2]
Pressure / viscous drag	~93% / ~7%	85–95% / 5–15%

PROJECT 04

SpaceX Dragon Capsule Re-Entry

Mach 20 · Blunt-body hypersonic re-entry

- Blunt-body re-entry geometry (GrabCAD, refined) to study bow-shock standoff and stagnation heating
- Laminar, density-based solve with NASA 9-piecewise Cp for high-temperature air
- Wall held at 1,000 K to examine the thermal protection boundary condition



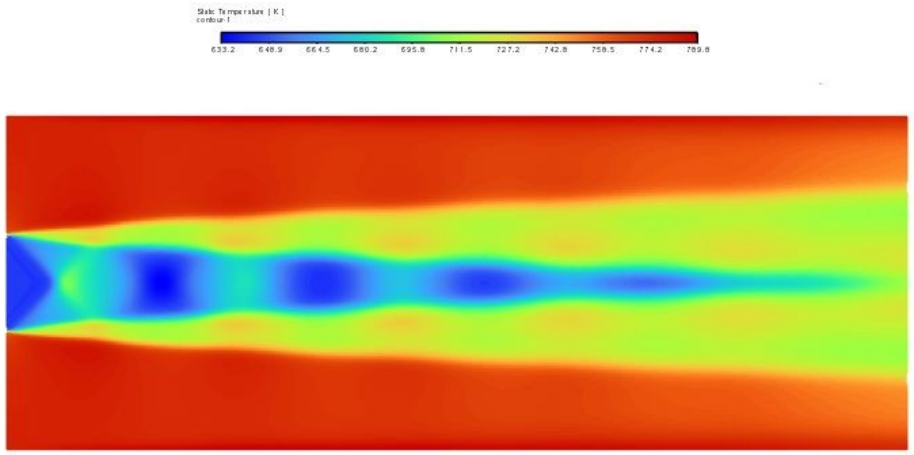
Freestream: 50 Pa, 216 K, Mach 20 · ~1.6M cells, 15-layer inflation

Parameter	CFD (this work)	Published
Drag coefficient (Cd)	1.26	1.23 [3]
Flow physics	Shock-driven	Shock-driven

PROJECT 05

LOX / LH_2 Rocket Combustion

Reacting flow · Pressure-based solver · H_2–O_2 species



- Simulated LOX/LH_2 mixing and combustion — the propellant pair used in NASA's high-performance engines
- Studied species mixing, reaction-zone formation and temperature rise downstream of the injector
- Radial-velocity contours reveal alternating vortical structures that enhance turbulent mixing

H_2: 0.0126 kg/s @ 858 K · O_2: 0.1 kg/s @ 1e6 Pa, 858 K · 17,160-element quad mesh

Efficient H_2–O_2 mixing proved to be the key driver of combustion intensity and temperature rise — underscoring how injector design and flow dynamics govern performance in cryogenic rocket engines.

LET’S BUILD SOMETHING FAST

EDUCATION

B.Tech in Mechanical Engineering

University College of Engineering, Narasaraopet — JNTUK

2016 – 2020 · CGPA 7.69

CONTACT

- LinkedIn — linkedin.com/in/veeresh-bandi-43b555211
- GitHub — github.com/sai-veeresh
- Based in Delhi, India

[1] Navó & Bergadà, “Aerodynamic Study of the NASA X-43A Hypersonic Aircraft,” Appl. Sci., 10(22), 2020. [2] Davis & White, J. Spacecr. Rockets, 45(3), 2008. [3] Gonzales et al., IEEE Aerospace Conf., 2014.